

SAR Lab 7: Xinmo Landslide Case Study

Pre-event InSAR deformation and post-event radar/optical change detection

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1 Introduction

This report investigates the Xinmo landslide in Sichuan Province, China, which occurred on 24 June 2017. The aim of the lab was twofold. First, I used pre-event Sentinel-1 interferometric SAR data to check whether the slope showed measurable deformation before the catastrophic failure. Second, I compared radar and optical satellite products before and after the event to evaluate which data source mapped the landslide most clearly.

The case is well suited for multi-sensor analysis because the area is steep, mountainous, and vegetated. These conditions make conventional radar coherence mapping difficult, but they also make vegetation and bare-ground changes very visible in optical indices. For that reason, the final interpretation combines InSAR time-series analysis, Sentinel-1 backscatter change, and Sentinel-2 spectral indices.

2 Data and Processing Workflow

The study area was clipped to the Xinmo landslide region, approximately between 103.62–103.695°E and 32.055–32.121°N. For the pre-event deformation analysis, I used public COMET-LiCSAR interferograms from frame 062D.05831.131313. Only interferograms ending before the landslide date were selected, with a maximum temporal baseline of 72 days. The LiCSAR unwrapped phase and coherence products were converted into MintPy-compatible HDF5 files using the supplied conversion workflow.

The main processing steps were:

1. Download pre-event LiCSAR unwrapped phase and coherence products for the Sentinel-1 descending frame.
2. Subset the data around the landslide using the geographic bounding box.
3. Convert the GeoTIFF interferograms into MintPy `ifgramStack.h5` and `geometryGeo.h5` files.
4. Run an SBAS time-series inversion in MintPy.
5. Select a local reference pixel outside the landslide area to reduce atmospheric and seasonal artifacts.
6. Compare pre- and post-event Sentinel-1 VV backscatter in Google Earth Engine.
7. Use Sentinel-2 optical images to compute NDVI, BSI, and NDWI changes.

The local reference point was selected outside the unstable slope, near the landslide but not inside the failure zone. This is important because InSAR displacement is relative: a poor reference pixel can introduce false trends into the target time series. In my processing, the reference point

was chosen in a nearby stable area, while the analyzed landslide pixel was located on the slope affected by the later collapse.

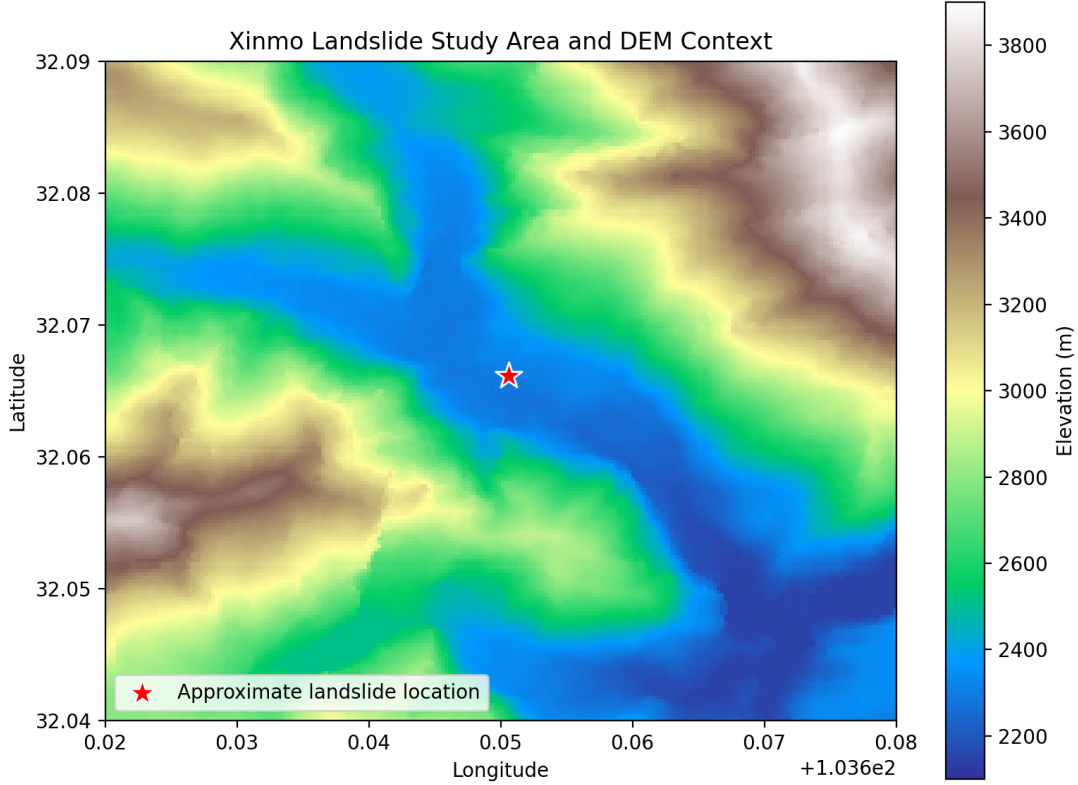


Figure 1: Topographic context and Area of Interest for the Xinmo landslide study.

3 Pre-event InSAR Time-series Results

The SBAS MintPy result shows that the Xinmo slope was not completely stable before the 2017 collapse. The selected landslide pixel has a negative line-of-sight displacement trend, meaning motion away from the Sentinel-1 sensor. In this geomorphic setting, that signal is consistent with slow downslope creep before the final landslide.

The representative time series indicates a gradual cumulative displacement of about 6 cm before the event. A linear estimate from the time series gives an approximate LOS velocity of -0.34 ± 0.05 cm/year. Although this is not an extremely rapid motion, it is persistent and coherent enough to support the interpretation of long-term gravitational slope deformation.

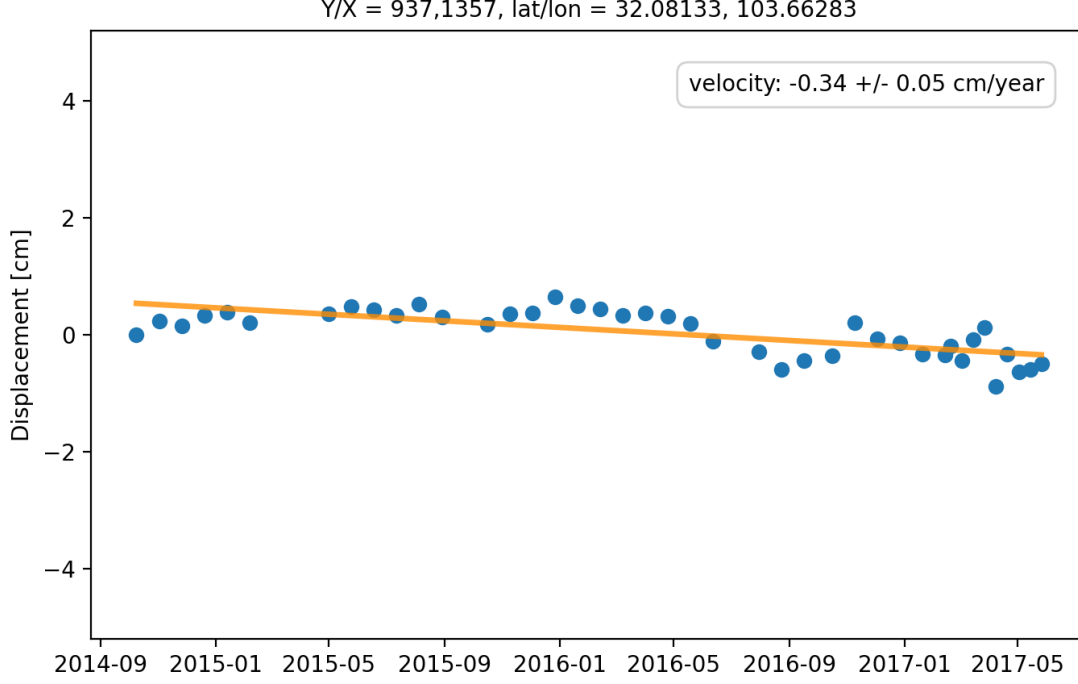


Figure 2: MintPy pre-event displacement time series for a representative landslide pixel.

One noticeable feature in the time series is a sharp drop and recovery around 2015. I interpret this as a processing artifact rather than real ground motion. The area is steep and vegetated, so temporal decorrelation, atmospheric delay, and phase unwrapping errors are expected. This single anomaly should not be over-interpreted, but the broader negative trend remains meaningful because it persists across the pre-event period.

4 Coherence and Radar Backscatter Change

The lab also asked whether coherence loss is useful for mapping the landslide. In this case, coherence is a weak mapping tool. The landslide area shows low coherence, but so do many surrounding slopes and valley areas. This is caused by dense vegetation, rugged topography, radar layover, and natural surface changes. As a result, coherence loss does not isolate the landslide boundary clearly.

Sentinel-1 VV backscatter change provides a more useful radar-based result. I compared a pre-event image from 12 June 2017 with a post-event image from 24 June 2017 and calculated the post-minus-pre VV difference. The individual backscatter images are noisy because of mountain radar geometry, but the difference image shows a more localized anomaly along the landslide scar and run-out zone. The positive VV change is consistent with rougher exposed debris and disturbed ground after the failure.

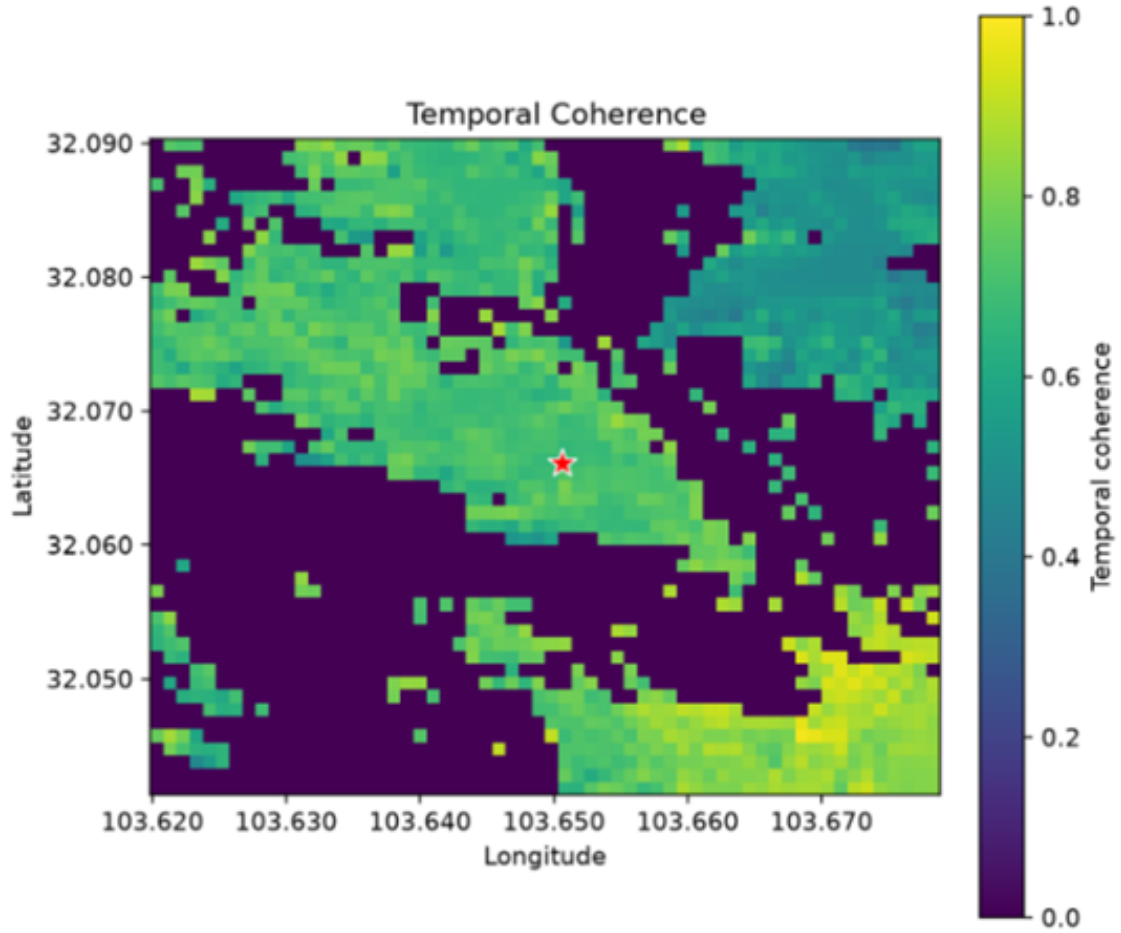


Figure 3: Coherence map for the interferogram covering the landslide event.

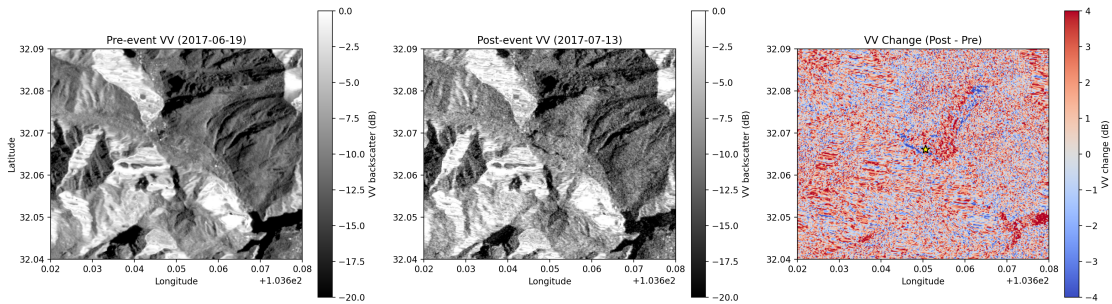


Figure 4: Sentinel-1 VV backscatter comparison before and after the landslide.

Overall, VV backscatter change is better than coherence loss for this site, but it is still not the cleanest way to map the landslide. Radar mapping is limited by the steep forested terrain, so the best radar contribution is a rough indication of where surface roughness changed, not a precise landslide outline.

5 Optical Change Detection

Sentinel-2 optical data produced the clearest post-event mapping result. I used cloud-free pre- and post-event imagery to calculate spectral index changes over the AOI. The most useful indices

were NDVI for vegetation loss, BSI for newly exposed bare soil and debris, and NDWI for moisture and drainage changes.

5.1 NDVI

NDVI is calculated from the near-infrared and red bands:

$$NDVI = \frac{NIR - Red}{NIR + Red}$$

Before the landslide, the affected slope had relatively high NDVI, showing that it was mostly vegetated. After the event, NDVI decreased strongly in the scar and run-out zone. The NDVI difference map shows a clear negative anomaly, with the largest decrease reaching approximately -0.5 . This indicates major vegetation removal and burial by landslide debris.

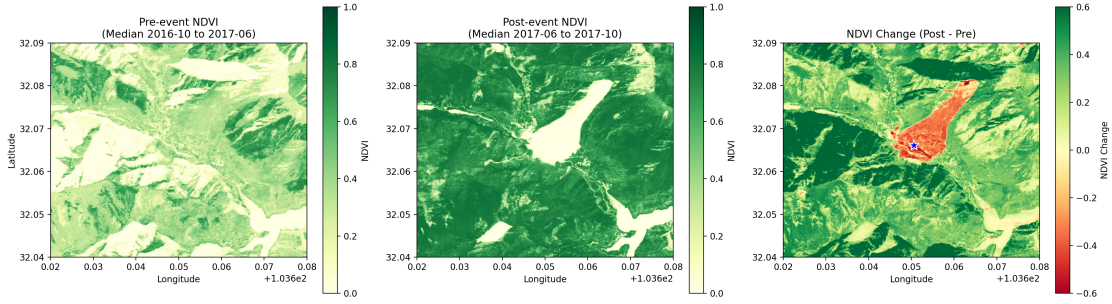


Figure 5: Sentinel-2 NDVI before, after, and after-minus-before change map.

5.2 BSI

The Barren Soil Index was used to highlight exposed soil, rock, and debris:

$$BSI = \frac{(SWIR + Red) - (NIR + Blue)}{(SWIR + Red) + (NIR + Blue)}$$

The post-event BSI map shows a strong increase over the landslide area. This agrees with the NDVI result: where vegetation disappeared, bare material became exposed. BSI is therefore a useful companion to NDVI because it confirms that the optical change is not only vegetation loss, but also the appearance of fresh soil, rock, and debris.

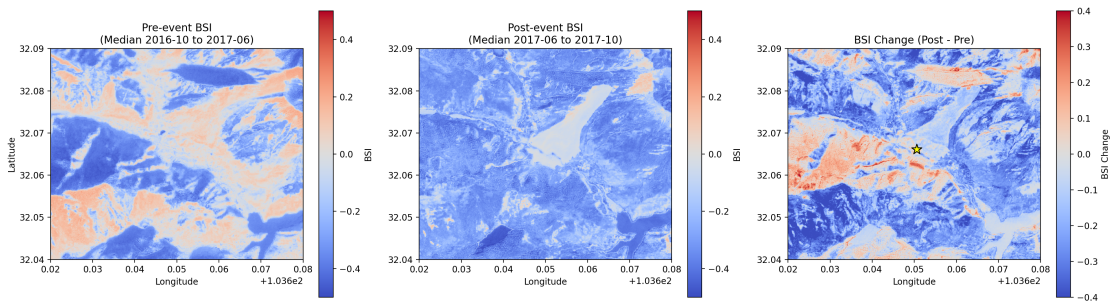


Figure 6: Sentinel-2 BSI before, after, and after-minus-before change map.

5.3 NDWI

NDWI was also useful for interpreting hydrological changes:

$$NDWI = \frac{Green - NIR}{Green + NIR}$$

The NDWI result suggests that the landslide changed the local drainage pattern. The upper source area became drier because vegetation and soil were removed, while parts of the lower run-out zone near the valley floor showed localized moisture changes. This is consistent with debris disturbing channels and changing how water moved through the valley after the collapse.

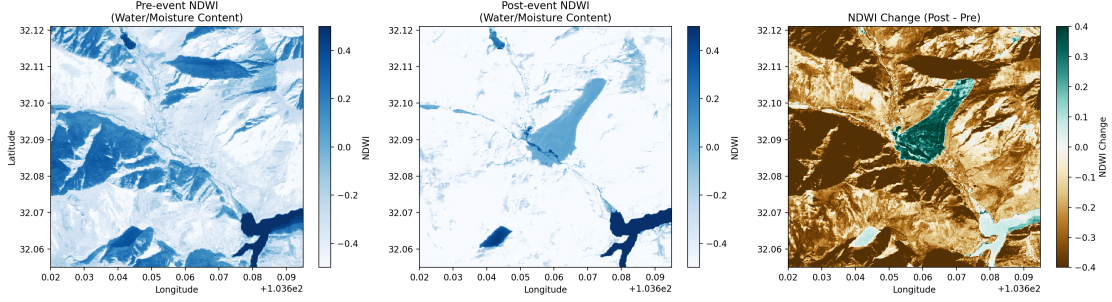


Figure 7: Sentinel-2 NDWI before, after, and change map.

6 Discussion

The different datasets answer different parts of the problem. The pre-event InSAR time series is most useful for detecting slope instability before failure. It shows a slow but persistent negative LOS trend, supporting the idea that the slope was creeping before the catastrophic collapse. However, InSAR is not the best tool for mapping the final landslide boundary after the event, especially when coherence is poor.

For post-event mapping, Sentinel-1 coherence loss performed poorly because decorrelation was already widespread across the steep vegetated terrain. VV backscatter change was better because it produced a more localized signal in the scar and run-out zone, but it remained noisy and affected by radar geometry. Sentinel-2 optical indices gave the most reliable landslide outline. NDVI clearly mapped vegetation removal, BSI mapped exposed debris and bare ground, and NDWI added information about moisture and drainage changes.

Table 1: Comparison of datasets used in the Xinmo landslide analysis.

Dataset / product	Main information	Usefulness
MintPy SBAS InSAR	Pre-event LOS deformation	Best for detecting slope creep
LiCSAR coherence	Phase stability / decorrelation	Weak for mapping this landslide
Sentinel-1 VV change	Radar roughness change	Moderate; localized but noisy
Sentinel-2 NDVI	Vegetation loss	Very strong mapping signal
Sentinel-2 BSI	Bare soil and debris exposure	Very strong mapping signal
Sentinel-2 NDWI	Moisture and drainage change	Useful supporting evidence

7 Conclusion

The Xinmo landslide area showed signs of instability before the 24 June 2017 failure. The MintPy SBAS time series from COMET-LiCSAR interferograms indicates slow negative LOS displacement, with about 6 cm of cumulative motion and an estimated velocity near -0.34 cm/year. This suggests long-term gravitational creep before the final collapse.

After the event, optical data mapped the landslide more clearly than radar coherence. Coherence loss was too widespread to isolate the landslide, while VV backscatter change provided a better but still noisy radar signal. The clearest results came from Sentinel-2 spectral indices: NDVI showed strong vegetation loss, BSI showed newly exposed soil and debris, and NDWI

helped identify hydrological disturbance. The strongest interpretation therefore comes from combining sensors: InSAR reveals pre-failure motion, while optical change detection maps the surface impact of the landslide.

Processing Commands

The LiCSAR download and MintPy conversion followed the supplied scripts:

```
python download_licsar_pre_event.py \  
    --base-url https://gws-access.jasmin.ac.uk/public/nceo_geohazards/\  
LiCSAR_products/62/062D_05831_131313/ \  
    --out licsar_062D_pre20170624 \  
    --cutoff 20170624 \  
    --max-days 72 \  
    --products unw cc png  
  
python licsar_to_mintpy_h5.py \  
    --licsar-dir licsar_062D_pre20170624 \  
    --out mintpy_xinmo \  
    --bbox 103.62 32.055 103.695 32.121  
  
smallbaselineApp.py smallbaselineApp_licsar.cfg --start modify_network
```